# RAWMER: REFERENCE-ASSISTED SPEECH WATERMARKING VIA RELATIVE MEL-ENERGY RELATIONS

*Chengxun Hong*

Author Affiliation(s)

## ABSTRACT

Providers of neural text-to-speech need to substantiate provenance claims about audio they generated, which motivates watermarking the intermediate mel spectrogram. Existing mel-domain schemes encode each bit as an absolute perturbation pattern and degrade sharply under strong additive noise. We propose `RAWMER`, which instead encodes each bit as the sign of an energy difference between two keyed groups of mel bins, repeated across time blocks. Because both groups lie in the same local time–frequency region, distortions affecting them equally—gain changes, mild filtering, vocoder bias—cancel in the difference; a selection step uses the stored clean reference to choose bin groups that avoid the dynamic-range boundary, carry stable energy and are balanced in reliability. Against our re-implementation of MelShield at matched clean quality (PESQ $\approx$ 3.51) on HiFi-GAN, `RAWMER` raises bit accuracy under additive noise at 10 and 5 dB SNR from 71.3% and 63.3% to 94.1% and 85.9%, with consistent gains on DiffWave. Further experiments show chance-level false positives, $\approx$ 22 KB references and verification from unaligned 25% clips. Time–frequency desynchronization remains the main limitation and the most likely adaptive attack.

***Index Terms***— Speech watermarking, provenance verification, mel spectrogram, neural text-to-speech, additive-noise robustness

## 1. INTRODUCTION

Modern text-to-speech (TTS) and voice cloning produce speech hard to distinguish from genuine recordings, raising risks of fraud, impersonation and misinformation [1] and motivating watermarks that trace audio to its source [2, 3]. We state up front which part of that problem we address. `RAWMER` is *provider-side*: only the generating provider, holding the secret key and a per-utterance reference, can verify, and only for audio it generated. It substantiates or refutes an ownership claim about a specific candidate generation record; it cannot flag a clone from a non-cooperating system, which remains the province of blind detection [2] and deepfake classifiers [1]—the two are complementary (§2.5).

**Related work.** Post-hoc waveform watermarks such as AudioSeal [2], WavMark [3] and SilentCipher [4] embed after synthesis and decode blindly. In-generation schemes tie the mark to the pipeline instead: Timbre Watermarking [5] into speaker timbre, TraceableSpeech [6] into discrete speech tokens, Groot [7] into a diffusion generator's noise prior. Since most TTS systems emit a mel spectrogram before a vocoder [8, 9], that mel is a shared, model-agnostic interface needing no vocoder retraining. MelShield [10] follows this route, embedding a keyed payload via low-energy spread-spectrum perturbations and verifying against a retained clean reference. We adopt the same setting and take MelShield—concurrent work, re-implemented locally—as our primary baseline.

Our point of departure is the encoding mechanism. MelShield [10] and prior mel-domain schemes represent each bit as an absolute perturbation pattern, detected by correlating the reference residual with the keyed pattern. Absolute residuals are sensitive to gain, frequency shaping and vocoder reconstruction error; at lower SNRs the pattern is buried and bit accuracy falls toward chance. This fragility under strong additive noise—not under compression or filtering—is our target.

Our remedy rests on a principle that is not new. Encoding a bit as the sign of an energy difference between two keyed sets of host samples is the classical *patchwork* construction [11, 12], one of a lineage of relational and quantization-based embeddings including QIM [13]; our detector is likewise an *informed* one [14], since it reads the host. Our claim is narrower, and has two parts. **Relative mel-energy encoding before vocoding:** moving patchwork-style differential encoding into the pre-vocoder mel domain yields a score $z_i$ in which common-mode distortion cancels; alone this already exceeds—narrowly—a quality-matched absolute-residual baseline under strong noise. **Reliability-aware bin-group selection:** the clean reference, already required here, picks among $C$ keyed candidates the pair whose groups both have dynamic-range headroom and stable energy and are balanced in reliability (Eq. (2)); this is the larger contributor. On HiFi-GAN [8] and DiffWave [9] the combination improves noise robustness over a quality-matched

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**Fig. 1**. RAWMER pipeline: (a) reference-guided bin-group selection among $C$ candidates; (b) differential encoding — the bit sets the sign of the group energy difference; (c) embed and verify — distortions common to both groups cancel in $z_i$. [Replace this caption with your own.]

MelShield [10] re-implementation, with chance-level false positives, $\approx 22$ KB references and verification from unaligned 25% clips. It does *not* withstand an attacker who desynchronizes time or frequency.

## 2. METHOD

`RAWMER` embeds an $N$-bit payload into the intermediate mel spectrogram before vocoding and recovers it from a suspect waveform using a stored clean mel reference (Fig. 1).

### 2.1. Setup

Let $X \in [0,1]^{F\times T}$ be the per-utterance min–max normalized log-mel and $X_B$ its watermark band of $B$ bins. We partition time into blocks of length $L$, stride $S$; each block $k$ writes a keyed subset $\mathcal{A}_k$ of $N_b$ payload bits, from a permutation seeded by $(\kappa, k)$, so every bit recurs across blocks.

### 2.2. Block-wise Relative Mel-Energy Encoding

For each bit $i \in \mathcal{A}_k$, a keyed SHA-256 draw on $(\kappa, \mathrm{uid}, i, k)$ selects $2P$ band bins forming a *bin-group pair*: disjoint positive and negative groups $\mathcal{P}^{\pm}_{i,k}$ of $P$ bins each (Fig. 1). The pair vector $p_{i,k} \in \mathbb{R}^B$ is $\pm 1/\sqrt{2P}$ on those groups and zero elsewhere. With bit polarity $s_i = 2m_i - 1$, a block-energy weight $w_k = \mu + (1-\mu)\bar{X}^{\gamma}_{B,k}$ that emphasizes voiced frames, and an active-bit scale $\eta = 1/\sqrt{N_b}$, we accumulate per-block updates into a layer $\ell \in \mathbb{R}^{B\times T}$ with $\ell[\,\cdot\,,t] = \sum_{k:\, t\in[t_k, t_k+L)} \sum_{i\in\mathcal{A}_k} s_i w_k \eta\, p_{i,k}$. A headroom mask $M = \mathrm{clip}\big(\min(X_B, 1-X_B)/\beta,\, 0,\, 1\big)$ suppresses updates near the dynamic-range boundary, and a reuse count $R[b,t] = \sum_{i,k} |p_{i,k}[b]| \mathbb{1}_{[t_k, t_k+L)}(t)$ spreads energy on revisited bins. The watermarked mel is

$$\tilde{X}_B = \mathrm{clip}\Big(X_B + \alpha\, M \odot \ell \,/\sqrt{\max(R,1)},\, 0,\, 1\Big), \qquad (1)$$

where $\alpha$ controls embedding strength. The provider stores $X_B$ (or a quantized subset, §3) as the reference, then vocodes $\tilde{X}$.

### 2.3. Reliability-Aware Pair Selection

A purely random $p_{i,k}$ may place either group on low-energy or near-boundary bins, where the perturbation is absorbed by $M$ or by vocoder reconstruction. For each $(i,k)$ we instead draw $C$ candidate pairs from independent keyed seeds and keep the one whose groups are both reliable. Let $r[b] = \mathrm{mean}_t\big[M[b,t](\mu{+}(1{-}\mu)X_B[b,t]^{\gamma})\big]$ over frames in the block, and let $\rho_j^{\pm}$ denote the mean of $r$ over the positive/negative group of the $j$-th candidate. We select

$$j^\star = \arg\max_j\ \min(\rho_j^+, \rho_j^-) - \lambda\,|\rho_j^+ - \rho_j^-|, \qquad (2)$$

with $\lambda = 0.25$: both groups must be reliable *and* balanced in reliability, avoiding pairs whose differential score would be dominated by one side. Selection costs no side information: the candidates are keyed, so the verifier re-derives $j^\star$ from the same reference. §3 isolates its contribution, which exceeds that of differential encoding alone.

### 2.4. Reference-Assisted Detection

Given a suspect waveform, we recompute the normalized log-mel from the reference statistics and align it by a bounded-shift search ($\pm\Delta_{\max}$ frames) maximizing Pearson correlation of frame-mean energy curves, compensating resampling or compression drift. After DC-removing $E = \tilde{X}'_B - X_B$ we re-derive the keyed pairs and accumulate, for each bit $i$ voted on by $V_i$ blocks,

$$z_i = \frac{1}{\sqrt{V_i}} \sum_{k:\, i \in \mathcal{A}_k} w_k \, \eta \, \langle E_k, \, p_{i,k} \rangle, \tag{3}$$

where $\langle E_k, p_{i,k} \rangle$ is the joint band–time mean of $E[:, t_k : t_k + L] \odot p_{i,k}$. We decode $\hat{m}_i = ⊮[z_i \geq 0]$ and verify when $\mathrm{ACC}(\hat{m}, m) \geq \tau$ ($\tau = 0.75$). Since $z_i$ is differential over two groups of the *same* local block, common-mode distortion—gain shifts, mild filtering, shared reconstruction bias—largely cancels, and balanced pairs keep zero-mean noise mean-zero in $\langle E_k, p_{i,k} \rangle$.

### 2.5. Threat Model

Following Kerckhoffs' principle we treat the algorithm as public; secrecy rests on the key $\kappa$ and the stored reference. Verification is *claim-checking*: the provider nominates a candidate record and we test it. Locating that record among many is a separate retrieval problem we do not solve—a miss is indistinguishable from our wrong-reference control—so claims are scoped to dispute resolution over a known candidate, and the reference database is secret material to protect at $\approx 22$ KB per utterance.

Every attack we evaluate is *non-adaptive*: signal processing by an attacker ignorant of $\kappa$. Two stronger adversaries bound the claims. If $\kappa$ leaks, the attacker recomputes every $(\mathcal{P}^{\pm}_{i,k})$, Eq. (2) included, and flips any bit by transferring energy between the groups—far weaker than the noise we test—so removal cost is set by key confidentiality, not embedding strength. If $\kappa$ *and* a reference leak, the attacker can forge a valid mark onto arbitrary audio and frame the provider. Lacking $\kappa$, an attacker can still desynchronize the signal, the effective attack per §3. Our claims are limited to non-adaptive distortion plus desynchronization.

## 3. EXPERIMENTS

### 3.1. Setup

We evaluate on LJSpeech [15] with HiFi-GAN [8] and DiffWave [9]: $F = 80$ mel bins, $n_{\text{fft}} = 1024$, 22.05 kHz, hop 256. RAWMER embeds 32 bits with $B = 40$ band bins ($20 : 60$), $L = S = 8$, $N_b = 6$, $P = 6$, $C = 16$, $\alpha = 0.435$ (HiFi-GAN) / 0.355 (DiffWave), $\Delta_{\max} = 12$, $\tau = 0.75$; code will be released. Attacks are MP3/AAC compression, $\pm 8\%$ scaling, 16 kHz resampling, 300–8000 Hz bandpass / 3 kHz lowpass filtering, additive Gaussian noise at 20/10/5 dB SNR, and 80 ms echo. ACC is the fraction of payload bits recovered; VR the fraction of utterances with ACC $\geq \tau$ (24 of 32 bits). PESQ and STOI are computed against the *unmarked vocoder output*, isolating embedding from synthesis cost; at the matched point clean STOI is 0.969 for both methods on HiFi-GAN (0.960 vs. 0.954 on DiffWave), so they match on intelligibility too. We use one single-speaker read-English corpus with ground-truth mels; end-to-end TTS mels, multi-speaker or expressive speech and re-recording are untested gaps we flag, not extrapolate over.

**Baselines.** We compare against (i) a re-implemented MelShield [10] and (ii) public AudioSeal [2] and WavMark [3] models as cross-paradigm waveform references (16 bits, post-vocoder). Our MelShield shares RAWMER's frontend, band, payload and $\Delta_{\max}$, uses mask floor 0.05, energy exponent 0.75, boundary margin 0.02, zero headroom and the calibrated $\tau = 0.61$ of [10]; only $\alpha$ is tuned to match clean PESQ (0.050 / 0.061). MelShield reports ACC 1.000 under non-noise attacks and 0.782/0.705 at 10/5 dB on HiFi-GAN, 0.779/0.701 on DiffWave, at published points of PESQ $\approx 4.13$ and 3.41 [10]. Ours reproduces that signature and, on DiffWave at comparable clean PESQ (3.515 vs. 3.408), is slightly *stronger* than published (0.823 at 10 dB), so it is not weakened; on HiFi-GAN our matched point gives 0.713 against a published 0.782 obtained in a higher-quality regime with a differently scaled $\alpha$ (0.25, band $20 : 55$). As $\alpha$ is not comparable across formulations, PESQ matching carries the whole comparability burden; an ACC-vs-PESQ curve over an $\alpha$ sweep would remove it.

### 3.2. Main Comparison

Table 1a reports robustness at matched clean quality. On HiFi-GAN, RAWMER lifts $\mathrm{ACC}_{\text{noise10}}$ from 71.3% to 94.1% and $\mathrm{ACC}_{\text{noise5}}$ from 63.3% to 85.9%; on DiffWave, from 82.3% to 92.2% and 72.2% to 83.5%. Non-noise attacks stay at ACC $\geq 0.983$ for all methods and are omitted for space. Estimates come from one sampled set; a three-seed check (2026–2028, random 500) varies by $\leq 0.002$ at 10 and 5 dB, far below those gaps. Lacking utterance-level confidence intervals, we treat small within-table gaps as insignificant.

The waveform rows are context, not baselines to beat: they decode *blindly* while our detector reads the exact clean reference, and informed detection is strictly easier [14]. AudioSeal keeps moderate bit accuracy but rejects most noisy samples (VR 0.14–0.30 at 20 dB, zero at 10/5 dB) and WavMark declines to decode $\geq 95\%$ of them—under heavy noise possibly correct blind behavior, not a defect.

**Matched threshold.** RAWMER's $\tau = 0.75$ is stricter than the $\tau = 0.61$ calibrated in [10], so one may ask whether that margin alone explains Table 1a; Table 1b reports VR under both. MelShield's VR depends heavily on $\tau$: on HiFi-GAN at $\tau = 0.75$ it collapses to 0.434 at 10 dB and 0.127 at 5 dB, against RAWMER's 0.991 and 0.918. The relaxed threshold is not free—at chance accuracy the binomial false-positive rate is 0.35% at $\tau = 0.75$ but 10.8% at $\tau = 0.61$—so that column is a bit-margin diagnostic, not a deployable operating point. We advocate $\tau = 0.75$ and use it for all controls below.

**Table 1**. LJSpeech [15], random 2000 (seed 2026); *nX* = additive Gaussian noise at $X$ dB SNR; "Ref." marks *informed* detectors, which read the clean reference. **(a)** Bit accuracy: mel-domain methods (`RAWMER`, MelShield [10]) carry 32 bits at matched clean PESQ $\approx 3.51$, while AudioSeal [2] and WavMark [3] (blind, forced-decoded) carry 16 bits on random 500—context, not matched competitors. **(b)** Verification rate at two thresholds (a clip verifies when ACC $\geq \tau$), with FPR the chance-level rate $\Pr[\mathrm{Bin}(32, \frac{1}{2}) \geq \lceil 32\tau \rceil]$: $\tau = 0.61$ buys VR at $31\times$ the FPR, so it is a bit-margin diagnostic only. Bold: best per column, ties included.

(a) bit accuracy

| Vocoder | Method | Ref. | none | n20 | n10 | n5 |
|---|---|---|---|---|---|---|
| HiFi-GAN | MelShield | ✓ | 0.999 | 0.903 | 0.713 | 0.633 |
| | `RAWMER` | ✓ | **1.000** | **0.994** | **0.941** | **0.859** |
| | AudioSeal | — | **1.000** | 0.974 | 0.790 | 0.647 |
| | WavMark | — | **1.000** | 0.752 | 0.511 | 0.493 |
| DiffWave | MelShield | ✓ | **1.000** | 0.969 | 0.823 | 0.722 |
| | `RAWMER` | ✓ | **1.000** | **0.990** | **0.922** | **0.835** |
| | AudioSeal | — | **1.000** | 0.966 | 0.754 | 0.618 |
| | WavMark | — | **1.000** | 0.670 | 0.512 | 0.498 |

(b) verification rate

| Vocoder | $\tau$ | FPR | Method | n20 | n10 | n5 |
|---|---|---|---|---|---|---|
| HiFi-GAN | 0.75 | 0.35% | MelShield | 0.969 | 0.434 | 0.127 |
| | | | `RAWMER` | **1.000** | **0.991** | **0.918** |
| | 0.61 | 10.8% | MelShield | 0.999 | 0.874 | 0.612 |
| | | | `RAWMER` | **1.000** | **1.000** | **0.995** |
| DiffWave | 0.75 | 0.35% | MelShield | 0.997 | 0.841 | 0.485 |
| | | | `RAWMER` | **1.000** | **0.983** | **0.871** |
| | 0.61 | 10.8% | MelShield | **1.000** | 0.973 | 0.872 |
| | | | `RAWMER` | **1.000** | **1.000** | **0.985** |

**Table 2**. `RAWMER` on HiFi-GAN [8]; *nX* = additive Gaussian noise at $X$ dB SNR. **(a)** Reference compression, ACC / VR at $\tau = 0.75$ (random 2000; uint4 band is random 500); "band" stores the 40-bin watermark band, "full" all 80 bins, and a float32 band-only reference would occupy 87.5 KB. **(b)** Candidate count $C$, ACC (random 500); $C = 1$ is differential encoding with a random bin-group pair, i.e. no selection. **(c)** Desynchronization, ACC / VR (random 500).

(a) reference compression

| Variant | KB | n20 | n10 | n5 |
|---|---|---|---|---|
| float32 (full) | 175.0 | 0.994 / 1.000 | 0.941 / 0.994 | 0.859 / 0.931 |
| uint8 (band) | 21.9 | 0.994 / 1.000 | 0.939 / 0.993 | 0.858 / 0.925 |
| uint4 (full) | 21.9 | 0.982 / 1.000 | 0.905 / 0.975 | 0.818 / 0.842 |
| uint4 (band) | 11.0 | 0.983 / 1.000 | 0.913 / 0.984 | 0.825 / 0.856 |

(b) candidates

| $C$ | PESQ | n10 | n5 |
|---|---|---|---|
| 1 | 3.55 | 0.767 | 0.668 |
| 4 | 3.53 | 0.889 | 0.785 |
| 8 | 3.52 | 0.923 | 0.832 |
| 16 | 3.51 | **0.944** | **0.861** |

(c) desynchronization

| Magnitude | pitch shift | speed |
|---|---|---|
| small | $\pm$25 c: 1.000 / 1.000 | 3%: 0.959 / 0.997 |
| medium | $\pm$50 c: 0.995 / 1.000 | 5%: 0.843 / 0.878 |
| large | $\pm$100 c: 0.784 / 0.728 | 10%: 0.675 / 0.290 |

### 3.3. Reference-Assisted Verification

**Negative controls.** On 2000 unattacked HiFi-GAN utterances with the correct reference, key and payload, `RAWMER` passes at VR $=$ 1.000; a wrong key, payload or reference gives ACC $\in [0.499, 0.503]$ and VR $\in [0.0015, 0.0040]$, matching the 0.35% binomial tail—the observed false positives are what chance alone predicts. Wrong-payload trials retain detector *confidence* on par with positives while their decoded bits disagree with the claim: verification must hinge on payload match, not confidence.

**Reference compression.** The stored reference is the band $X_B$, not the full mel. Table 2a shows 8-bit quantization of $X_B$ shrinks it to about 22 KB while ACC drops by at most 0.0013—$1/8$ of the float32 full-mel row (175.0 KB), or $1/4$ of the tighter float32 band-only baseline (87.5 KB). The same 22 KB on a 4-bit full-mel reference is consistently worse: precision inside the band dominates coverage outside it. Under band-only storage our MelShield re-implementation degrades far further (0.639/0.577/0.550 at 20/10/5 dB versus 0.902/0.713/0.631 with a full float32 reference, random 500), as expected of an absolute-residual detector.

**Blind partial-clip verification.** Because each bit repeats across keyed blocks, `RAWMER` survives cropping. Without revealing the crop offset, a 25% slice from the start, middle or end is slid across the reference (stride 4 frames) and the highest-scoring window taken: median start error 0–1 mel frames, ACC 0.985–0.991, VR $\geq$ 0.998 at all three positions; 50% clips reach 1.000.

### 3.4. Ablation and Robustness Boundary

**Which mechanism matters.** Table 2b sweeps $C$: going from $C = 1$ to 16 lifts $\mathrm{ACC}_{\mathrm{noise5}}$ from 66.8% to 86.1% at negligible PESQ cost, isolating Eq. (2) as the dominant contributor. Read honestly, differential encoding *alone* ($C = 1$, random pairs) beats quality-matched MelShield at 5 dB, 66.8% vs. 63.3%, but narrowly; selection supplies the remaining 19.3 points. We have not run the converse ablation—absolute-residual encoding *with* reliability-aware placement—so the differential score's independent value rests on $C = 1$ alone.

**Robustness boundary.** `RAWMER` withstands 64 kbps MP3, 48 kbps AAC, 2 kHz lowpass, 8-bit quantization, $\pm$0.5 clipping, 100 Hz–7 kHz bandpass and stronger reverb (ACC $\geq$ 0.987 on both vocoders), plus EnCodec [16] at 6/12/24 kbps (ACC $\geq$ 0.976 versus 0.888/0.942/0.954 for quality-matched MelShield on HiFi-GAN): the advantage is not confined to Gaussian noise, though noise is where the methods separate most. At 0 dB SNR it degrades gracefully (ACC 0.73–0.75, VR $\leq$ 0.60).

**Desynchronization is the attack to beat.** The genuine limit is time–frequency desynchronization (Table 2c). It arises from the bounded-shift alignment of §2.4 breaking down, not from watermark energy being lost—which is why it is dangerous. A keyless attacker who knows only that alignment is frame-level can apply a $\pm 100$-cent shift or $\pm 10\%$ speed change, both trivially scriptable and leaving speech intelligible, and will prefer this to additive noise. Two keyless attacks are quantified in the supplement: [band-concentrated noise results TBD] and [re-vocoding results TBD]. Restoring synchronization (multi-scale search, DTW) is thus a security requirement.

## 4. CONCLUSION

`RAWMER` carries payload bits in block-wise relative mel-energy relations between reliability-selected bin groups: the classical patchwork principle [11, 12] moved before the vocoder, with the already-required clean reference spent on choosing which groups to use. It is scoped to provider-side claim checking, and synchronization remains the open problem.

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